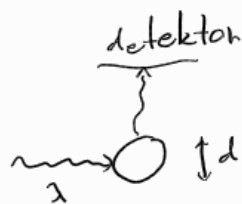


Velikost jedra

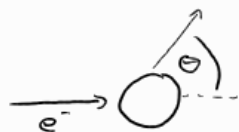
• Rutherfordov poskus 1911: $r_{jed} \ll r_{at.}$

oznake jader: ${}^A_Z X$



$\rightarrow \lambda \gg d$: sipanje izotropno

valovanja: $\gamma, e, n, p, \dots$



$$\frac{\Delta N}{\Delta \Omega}(\theta) \rightarrow \frac{\partial \sigma}{\partial \Omega}(\theta)$$

← diferencijalni sipalni presjek
 $\hookrightarrow$ enota: $\frac{m^2}{sr} = m^2$
 ← steradian

Dif. sip. presjek

$$d = \lambda = 2r_j$$

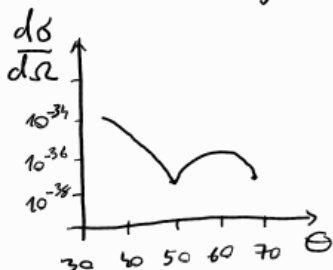
$$p = \frac{h}{\lambda} = \frac{hc}{d} = \frac{2\pi hc}{d} \approx 600 \text{ MeV/c}$$

$$hc = 200 \text{ MeV fm} = 200 \text{ eV nm}$$

$$E = \sqrt{p^2 c^2 + m^2 c^4} \approx 600 \text{ MeV}$$

$\downarrow$
9,5 MeV za e^-

Primer: sipanje e^- z $E = 420 \text{ MeV}$ na ${}^{12}\text{C}$



Uklon na okrogli reži (iz RF):

$$1. \text{ minimum } \sin \Theta = 1,22 \frac{\lambda}{d} \rightarrow d = 1,22 \frac{\lambda}{\sin \Theta} \approx 2 \text{ fm}$$

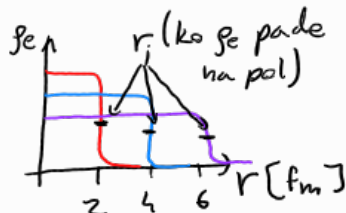
Če ponovimo na razl. jedrih: $r_j = A^{1/3} r_0$; $r_0 = 1,1 \text{ fm}$

$$\rightarrow V_j \propto A$$

Porazl. naboja v jedru

višje $E e^- \Rightarrow \lambda < d$

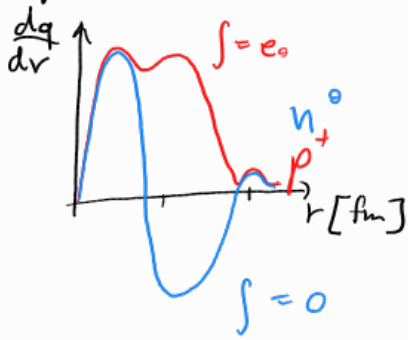
$$\frac{d\sigma}{d\Omega} \rightarrow f_e(r)$$



• Ca
 • Sr
 • Au

$$f_e = \frac{f_0}{e^{-\left(\frac{r-r_j}{s}\right)} + 1}$$

porazdelitev naboja v protonu in nevtronu



$$\frac{dQ}{dr} = \rho_e 4\pi r^2$$

↓
delež naboja v lupini debeline dr

Masa jedra

$$\rho = eBR$$

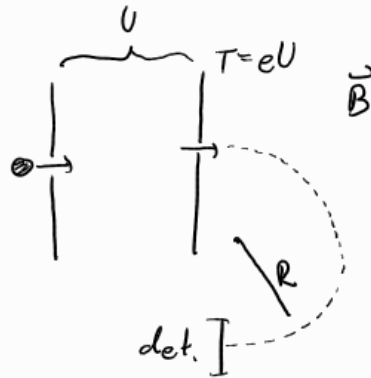
Masni spektrometer

$$T \ll m_j c^2 \Rightarrow T = \frac{1}{2} m_j v^2$$

$$eU = \frac{1}{2} m_j v^2; m_j v = eBR$$

$$\downarrow$$

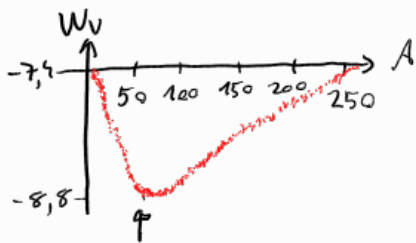
$$m_j \sqrt{\frac{2eU}{m_j}} = eBR \rightarrow \boxed{m_j = \frac{e^2 B^2 R^2}{2eU}}$$



$m_j < Zm_p + Nm_n$ pričakujemo, ker je drugače nevezano stanje energijsko bolj ugodno

$$W_v = m_j c^2 - Zm_p c^2 - Nm_n c^2 < 0 \quad \text{Vezavna energija}$$

$$W_v = \frac{W_v}{A} \quad \text{spec. vez. en.} \quad -8,7 \text{ MeV} < W_v < -7,4 \text{ MeV}$$

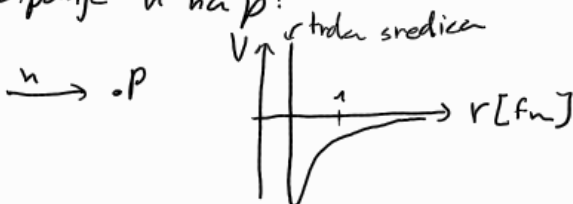


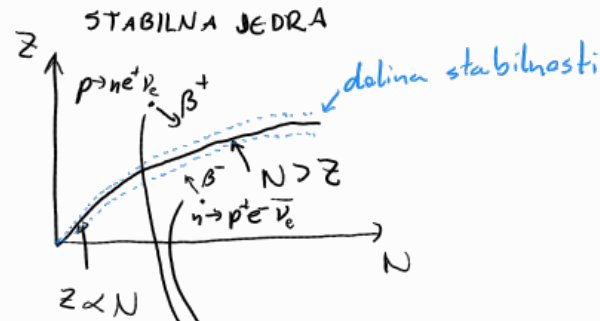
minimum: $A=56$

${}^{56}\text{Fe}$ najbolj vezano jedro

• nukleoni čuti samo privlak sosedu $\rightarrow$ jedrska sila je kratkega dosegu

• sipanje n na p:





če ima jedro preveč p/h , se prek β^+ razpadov prenakne v dolino stabilnosti

Semiempirična masna formula

$$W_V(A, Z) = -w_0 A + w_1 A^{2/3} + w_2 \frac{Z^2}{A^{1/3}} + w_3 \frac{(A-2Z)^2}{A} + w_4 \frac{\sigma_{ZN}}{A^{3/4}}$$

$\downarrow$ 15,6 MeV $\downarrow$ 17,3 MeV $\downarrow$ 0,70 MeV $\downarrow$ 23,3 MeV $\downarrow$ 33,5 MeV

$$\sigma_{ZN} = \begin{cases} -1; & Z, N \text{ soda} \\ 0; & Z \text{ lih, } N \text{ sod} / Z \text{ sod, } N \text{ lih} \\ 1; & Z, N \text{ liha} \end{cases}$$

Vrtilna količina jedra in magnetni moment

$$\vec{J} = \vec{L} + \vec{S}$$

atom: $\mu_B = \frac{e \hbar}{2 m_e}$

jedro: $\mu_j = \frac{e \hbar}{2 m_p} = \frac{m_e}{m_p} \mu_B = 5,05 \cdot 10^{-27} A m^2 \doteq 3 \cdot 10^{-8} \frac{eV}{T}$

e^- : $\mu_z = -g m_s \mu_B$; $g = 2$ giromagnetno razmerje

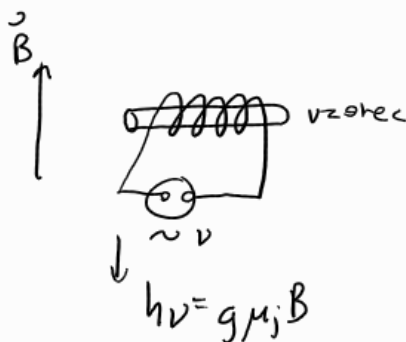
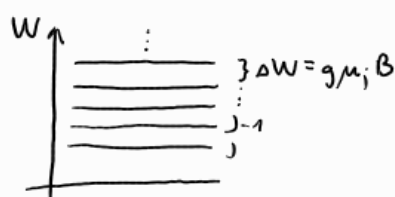
p^+ : $\mu_z = g_p m_s \mu_j$; $g_p = 5,586 \rightarrow \mu_z(m_s = \frac{1}{2}) = 2,793 \mu_j$

n^0 : $\mu_z = g_n m_s \mu_j$; $g_n = -3,826 \rightarrow \mu_z(m_s = \frac{1}{2}) = -1,913 \mu_j$

$$(\mu_z)_{M_j} = g M_j \mu_j$$

→ kako meriti g ?

$\vec{B} = (0, 0, B)$; $W = -\vec{\mu} \cdot \vec{B} = -\mu_z B = -g M_j \mu_j B$; $M_j = -J, -J+1, \dots, J$

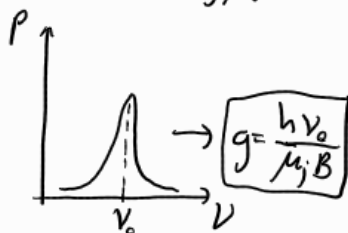


klasična slika

$$\vec{B} \downarrow \vec{\mu} \quad \vec{M} = \vec{\mu} \times \vec{B}$$

→ $\vec{\mu}$ se zasuka, M_z se spremeni

→ $(\mu_z)_{M_j} = g M_j \mu_j$ magnetni moment jedra

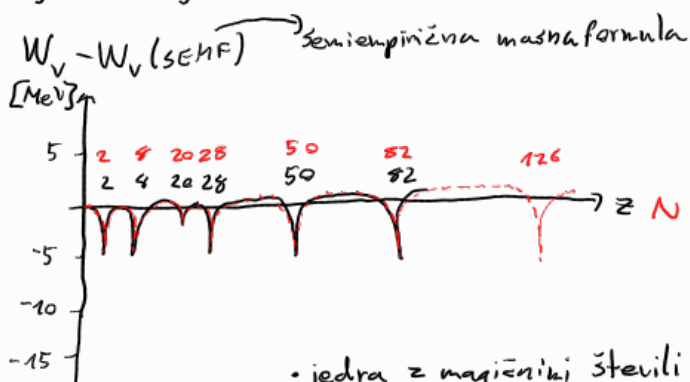


za $p = {}^1\text{H}$: $g = g_p = 5,586$, $J = S = \frac{1}{2}$
 $\rightarrow (\mu_z)_{m_j = J} = g_p J \mu_B \rightarrow \mu_B = \frac{g_p \mu_B}{h} \approx 42,6 \text{ MHz} \rightarrow \lambda = 7 \text{ m}$

Uporaba: jedrska magnetna resonanca (NMR):

- struktura snovi (npr. določena v neko polje in opazujemo resonanco)
- kemijski premik
- medicinsko slikanje (MRI)
- gostota jeder $F(\vec{r})$
- ${}^1\text{H}$, ${}^{13}\text{C}$, ${}^{15}\text{N}$, ${}^{19}\text{F}$, ${}^{31}\text{P}$

Zgradba jeder



- jedra z magičnimi števili n ali p boljše vezana

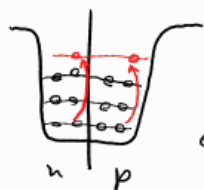
$$2, 8, 20, 28, 50, 82, 126$$

- več stabilnih jeder z mag. št. n ali p

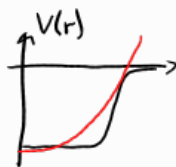
npr. ${}_{82}^{208}\text{Pb}$ ${}_{126}^{126}$ dvojno-magično št.

- jedra z N ali $Z = \text{mag. št.} + 1$ slabše vezana - bolj reaktivna (npr. K)

Nukleon v jedru čuti efektivni potencial



• če bi čutil vsak delec posebej, bi ob interakciji potrebovali veliko energije, da preideta v zg. en. stanje



$$V_{\text{ef}} = -\frac{V_0}{e^{\frac{r-r_0}{a}} + 1}$$

Woods-Saxonov potencial

$$-\frac{\hbar^2}{2m} \nabla^2 R(\vec{r}) + V_{\text{ef}}(r) R(\vec{r}) = W R(\vec{r})$$

Rešitev: numerično

približek W-S potenciala: parabola

$$V_{\text{ef}}(r) = \frac{1}{2} m \omega^2 r^2$$

$$\rightarrow -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) R(x, y, z) + \frac{1}{2} m \omega^2 (x^2 + y^2 + z^2) R(x, y, z) = W R(x, y, z) \rightarrow \text{postavka za } R:$$

$$-\frac{\hbar^2}{2m} (X''YZ + XY''Z + XYZ'') + \frac{1}{2} m \omega^2 (x^2 + y^2 + z^2) XYZ = WXYZ \quad R(x, y, z) = X(x)Y(y)Z(z)$$

$$\left(-\frac{\hbar^2}{2m} X''YZ + \frac{1}{2} m \omega^2 x^2 XYZ \right) + \left(-\frac{\hbar^2}{2m} XY''Z + \frac{1}{2} m \omega^2 y^2 XYZ \right) + \left(-\frac{\hbar^2}{2m} XYZ'' + \frac{1}{2} m \omega^2 z^2 XYZ \right) = WXYZ$$

$$\underbrace{\left(-\frac{\hbar^2}{2m} \frac{X''}{X} + \frac{1}{2} m \omega^2 X^2\right)}_{=W_x = \text{konst.}} + \underbrace{\left(-\frac{\hbar^2}{2m} \frac{Y''}{Y} + \frac{1}{2} m \omega^2 Y^2\right)}_{=W_y = \text{konst.}} + \underbrace{\left(-\frac{\hbar^2}{2m} \frac{Z''}{Z} + \frac{1}{2} m \omega^2 Z^2\right)}_{=W_z = \text{konst.}} = W$$

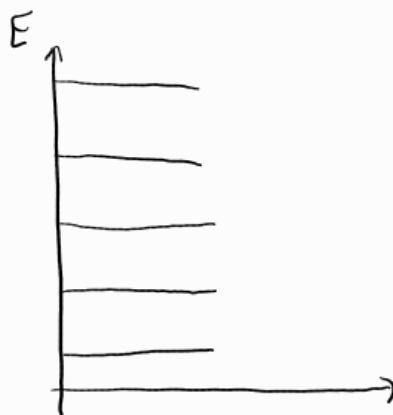
$$-\frac{\hbar^2}{2m} X'' + \frac{1}{2} m \omega^2 X^2 = W_x X \quad \text{harm. osc.} \rightarrow W_x = \hbar \omega \left(\frac{1}{2} + n_x\right); n_x = 0, 1, 2, 3, \dots$$

$$\dots \rightarrow W_y = \hbar \omega \left(\frac{1}{2} + n_y\right); n_y = 0, 1, \dots$$

$$\dots \rightarrow W_z = \hbar \omega \left(\frac{1}{2} + n_z\right); n_z = 0, 1, \dots$$

$$\rightarrow W_n = \hbar \omega \left(\frac{3}{2} + n_x + n_y + n_z\right)$$

n	E_n	n_x	n_y	n_z	3t. stanj (x2 zar. spina)	Σ do sedaj
0	$\frac{3}{2} \hbar \omega$	0	0	0	2	2
1	$\frac{5}{2} \hbar \omega$	1	0	0	6	8
		0	1	0		
		0	0	1		
2	$\frac{7}{2} \hbar \omega$	2	0	0	12	20
		0	2	0		
		0	0	2		
		1	1	0		
		1	0	1		
		0	1	1		
3	$\frac{9}{2} \hbar \omega$	3	0	0	20	40
		0	3	0		
		0	0	3		
		2	1	0		
		2	0	1		
		1	2	0		
		1	0	2		
		0	2	1		
		0	1	2		
		1	1	1		
4	$\frac{11}{2} \hbar \omega$				30	70
5	$\frac{13}{2} \hbar \omega$				42	112



Harmonski oscilator:

2, 8, 20, 40, 70, 112, ...

Magična števila:

2, 8, 20, 28, 50, 82, 126, ...

$\Rightarrow$ HO je pregrab približek
(ne dobimo pravih MŠ)

$$R(\vec{r}) = X(x) Y(y) Z(z)$$

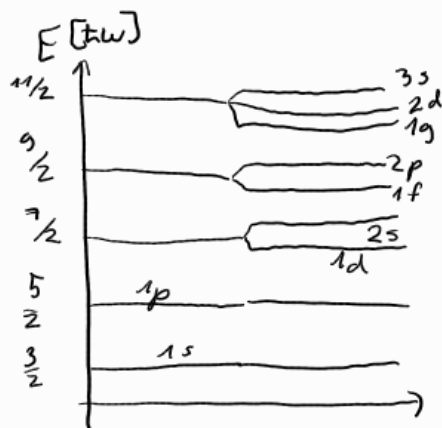
$$\rightarrow R(\vec{r}) = R(r) Y_{lm}(\vartheta, \varphi)$$

$$n = 2n_r + l + 1$$

$$E_n = \hbar \omega \left(\frac{1}{2} + n \right)$$

n	n _r	l	#	oznaka
1	0	0	2	1s
2	0	1	6*	1p
3	1	0	12	2s
	0	2		1d
4	0	3	20	1f
	1	1		2p
5	0	4	30	1g
	2	0		2d
	1	2		3s

l	oznaka
0	s
1	p
2	d
3	f
4	g
5	h
6	i



* #stanj: $2l+1$

pričakujemo stanje z višjim l
bo bolje vezano

$$H \rightarrow H - C \frac{dV}{dr} \vec{l} \cdot \vec{s}$$

$$|l, l_z, s, s_z\rangle \rightarrow |j, j_z, l, s\rangle$$

dobra k.s.

$$\rightarrow E \rightarrow E + \langle j, j_z, l, s | \underbrace{\left(-C \frac{dV}{dr} \right)}_{C'} \hat{\vec{l}} \cdot \hat{\vec{s}} | j, j_z, l, s \rangle$$

$$\langle j, j_z, l, s | \hat{\vec{l}} \cdot \hat{\vec{s}} | j, j_z, l, s \rangle = \langle j, j_z, l, s | \frac{1}{2} (\hat{j}^2 - \hat{l}^2 - \hat{s}^2) | j, j_z, l, s \rangle = \frac{1}{2} (j(j+1) - l(l+1) - s(s+1))$$

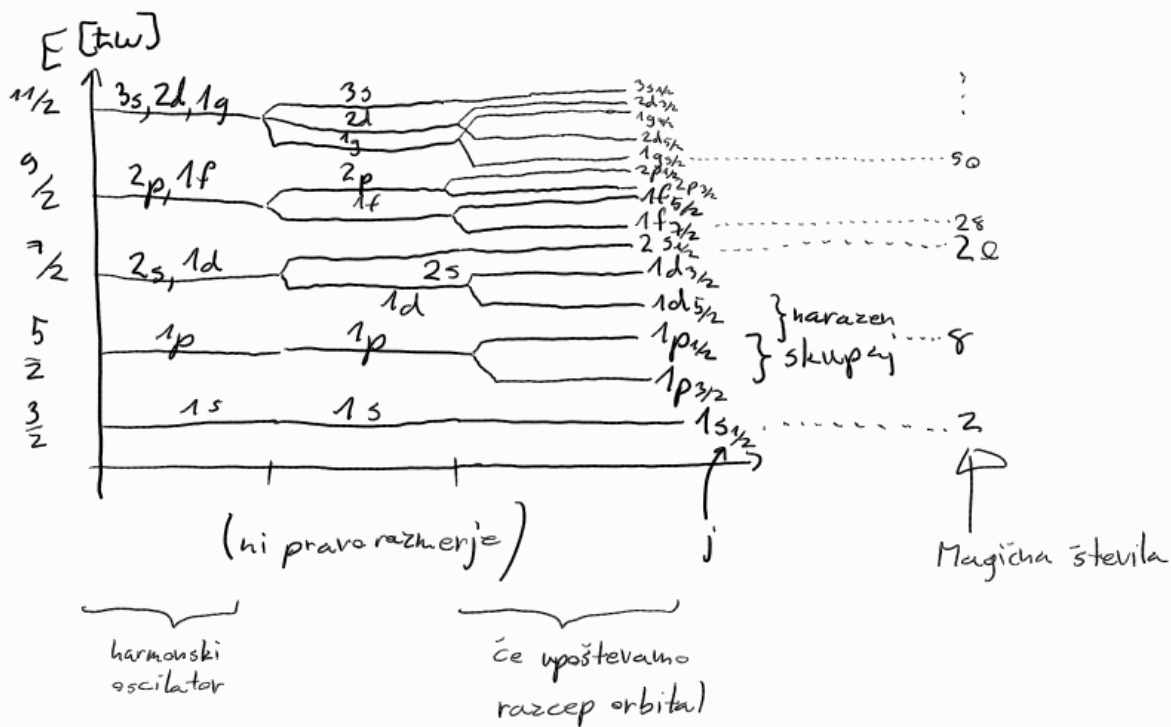
$$\hat{\vec{l}} \cdot \hat{\vec{s}} = \frac{1}{2} (\hat{j}^2 - \hat{l}^2 - \hat{s}^2)$$

$$1) j = l + \frac{1}{2}$$

$$2) j = l - \frac{1}{2}$$

$$\frac{1}{2} \left((l + \frac{1}{2})(l + \frac{3}{2}) - (l(l+1) - \frac{3}{4}) \right) = \frac{1}{2} l \quad \frac{1}{2} \left((l - \frac{1}{2})(l + \frac{1}{2}) - (l(l+1) - \frac{3}{4}) \right) = \frac{1}{2} (-l - 1) = -\frac{1}{2} (l+1)$$

$$\rightarrow E = C' \cdot \begin{cases} \frac{1}{2} l; j = l + \frac{1}{2} \\ -\frac{1}{2} (l+1); j = l - \frac{1}{2} \end{cases} \rightarrow \Delta E = E(l + \frac{1}{2}) - E(l - \frac{1}{2}) \propto - (l + \frac{1}{2})$$



• podlupine se polnijo ločeno za p^+ in n^0

• p in n se paroma sklopijo v VK 0

• primer: ^5Li : $\left(\overset{p}{(1s_{1/2})^2} (1p_{3/2})^1 \right) \left(\overset{n}{(1s_{1/2})^2} \right) \rightarrow J = \frac{3}{2}$

Parnost: $\hat{P}: \vec{r} \rightarrow -\vec{r}$

$$\hat{P}\psi = \begin{cases} +\psi; & \text{soda } (P=1) \\ -\psi; & \text{liha } (P=-1) \end{cases}$$

$$P(Y_{lm}(\theta, \varphi)) = (-1)^l$$

• primer: ^5Li : $l=1 \Rightarrow P=-1$ oznaka: $J^P = \frac{3}{2}^-$

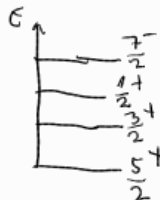
^{17}O : edini, ki prihaja VK, je n v stanju $1d_{5/2} \rightarrow J = \frac{5}{2}$, $l=2 \rightarrow J^P = \frac{5}{2}^+$

VZBUJENA STANJA JEDER

1) enodelčna vzb. stanja

^{17}O : ni $J = \frac{5}{2}^+ \rightarrow 1d_{5/2}, 2s_{1/2}, \dots$

tipične E : MeV



2) rotacijska vzb. stanja

$$W = \frac{\hbar^2 J(J+1)}{2I} \quad \left(\text{klasično } \frac{L^2}{2J} \right)$$

↑
vrt. moment

• samo za elipsoidna jedra, $J \neq 0$ (jedra, ki so daleč od mag. št.)

